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Lab Manual 8

A transportation company maintains records of its vehicles. Every vehicle stores:

- **Vehicle ID**
- **Model name**
- **Fuel level**

There are two types of vehicles:

1. **ElectricVehicle**
   - **Has a battery capacity**
   - **Can calculate remaining range based on battery level**
2. **FuelVehicle**
   - **Has a fuel tank capacity**
   - **Can calculate distance possible based on fuel consumption rate**

The company also employs a SystemInspector who needs access to sensitive internal data of all vehicles (including battery capacity and fuel tank details). However, these details must remain private and not directly accessible outside the class hierarchy

Code:

```
#include <iostream>
#include <string>

using namespace std;

class SystemInspector;

class Vehicle {
private:
    string vehicleID;
    string modelName;

protected:
    double fuelLevel;
```

```cpp
    friend class SystemInspector;

public:
    Vehicle(string id, string model, double fuel)
        : vehicleID(id), modelName(model), fuelLevel(fuel) {}

    virtual ~Vehicle() = default;

    virtual void displayInfo() const {
        cout << "Vehicle: " << modelName << " | ID: " << vehicleID << "\n";
    }
};

class ElectricVehicle : public Vehicle {
private:
    double batteryCapacity;

    friend class SystemInspector;

public:
    ElectricVehicle(string id, string model, double currentCharge, double capacity)
        : Vehicle(id, model, currentCharge), batteryCapacity(capacity) {}

    double calculateRemainingRange(double rangePerUnit) const {
        return fuelLevel * rangePerUnit;
    }
};

class FuelVehicle : public Vehicle {
private:
    double fuelTankCapacity;

    friend class SystemInspector;

public:
    FuelVehicle(string id, string model, double currentFuel, double capacity)
        : Vehicle(id, model, currentFuel), fuelTankCapacity(capacity) {}

    double calculateDistancePossible(double fuelEfficiency) const {
        return fuelLevel * fuelEfficiency;
    }
};

class SystemInspector {
```

```
public:
    void inspectElectricVehicle(const ElectricVehicle& ev) const {
        cout << "--- EV INSPECTION ---\n";
        cout << "ID: " << ev.vehicleID << "\n";
        cout << "Current Charge: " << ev.fuelLevel << " kWh\n";
        cout << "Battery Capacity: " << ev.batteryCapacity << " kWh\n\n";
    }

    void inspectFuelVehicle(const FuelVehicle& fv) const {
        cout << "--- FUEL VEHICLE INSPECTION ---\n";
        cout << "ID: " << fv.vehicleID << "\n";
        cout << "Current Fuel: " << fv.fuelLevel << " L\n";
        cout << "Tank Capacity: " << fv.fuelTankCapacity << " L\n\n";
    }
};

int main() {
    ElectricVehicle ev("EV-772", "Tesla Model 3", 50.5, 75.0);
    FuelVehicle fv("FV-109", "Toyota Corolla", 30.0, 50.0);

    ev.displayInfo();
    cout << "Remaining Range: " << ev.calculateRemainingRange(6.5) << " km\n\n";

    fv.displayInfo();
    cout << "Possible Distance: " << fv.calculateDistancePossible(15.0) << " km\n\n";

    SystemInspector inspector;

    inspector.inspectElectricVehicle(ev);
    inspector.inspectFuelVehicle(fv);

    return 0;
}
```

Output:

Compile Result

```
Vehicle: Tesla Model 3 | ID: EV-772
Remaining Range: 328.25 km

Vehicle: Toyota Corolla | ID: FV-109
Possible Distance: 450 km

--- EV INSPECTION ---
ID: EV-772
Current Charge: 50.5 kWh
Battery Capacity: 75 kWh

--- FUEL VEHICLE INSPECTION ---
ID: FV-109
Current Fuel: 30 L
Tank Capacity: 50 L


[Process completed - press Enter]
```